International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2025)

End-Semester Exam

Time: 180 Minutes　　　　Total Marks: 50

General Guidelines
• Attempt **any five (5) questions only**. • Unless stated otherwise, standard theorems from class may be used. • Show **all steps clearly**; unsupported answers may not receive full credit.

Question 1 [10 Marks]. Let $L, M, P \in \mathbb{R}$. Let (x_n) be a real sequence satisfying

(P1) $\lim_{n\to\infty}(x_{n+2} - x_n) = M$, (P2) $\lim_{n\to\infty}(x_{n+1} + x_n) = 2L$, and

(P3) $\lim_{n\to\infty}(x_{n+1} - x_n) = P$.

Find relation between M, L, and P such that the conditions are compatible. Then prove that the sequence (x_n) is convergent and converges to L.

Question 2 [10 Marks]. Answer the following questions.

(A). [**5 Marks**] Prove that the limit of a converging sequence is unique.

(B). [**5 Marks**] Let $\sum u_n$ be a series of positive real numbers and it is given that

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = L,$$

where L is a real number. Show that the series is convergent if $L > 1$.

Question 3 [10 Marks]. Let $(a_n)_{n\geq 1}$ be a real sequence. Let $b_n = \frac{a_n+a_{n+1}+a_{n+2}}{3}$ for all $n \geq 1$. Let the following two series converge.

$$E := \sum_{n=1}^{\infty} a_{2n}, \quad B := \sum_{n=1}^{\infty} b_{2n}.$$

Prove that the series $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} a_n = 3B - E + a_1 + a_2$.

Question 4 [10 Marks]. Let $X \subset \mathbb{R}$. Consider the following two statements. (a). X is closed and bounded. (b). Every sequence $(a_n)_{n=0}^{\infty}$ in X has a subsequence $(a_{n_j})_{j=0}^{\infty}$ which converges to a point $L \in X$.

(A). [**5 Marks**] $(a) \Rightarrow (b)$.

(B). [**5 Marks**] $(b) \Rightarrow (a)$.

Question 5 [10 Marks]. Let $X \subseteq \mathbb{R}$ and let $x_0 \in \mathbb{R}$ be a *limit point* of X. Let $f : X \to \mathbb{R}$ be a function, and let $L \in \mathbb{R}$. Consider the following two statements. (a) $\lim_{x\to x_0;\ x\in X} f(x) = L$. (b) For every sequence $(a_n)_{n=0}^{\infty} \subseteq X$ such that $\lim_{n\to\infty} a_n = x_0$, we have $\lim_{n\to\infty} f(a_n) = L$. Prove that

(A). [**5 Marks**] $(a) \Rightarrow (b)$.

(B). [**5 Marks**] $(b) \Rightarrow (a)$.

Question 6 [10 Marks]. Let $X \subseteq \mathbb{R}$ and let $c \in \mathbb{R}$ be a *limit point* of X. Let $f, g, h : X \to \mathbb{R}$ be three functions and let $L, E \in \mathbb{R}$.

(A). [**5 Marks**] If $\lim_{x\to c;\ x\in X} f(x) = L$. Then, for $n \in \mathbb{N}$, prove that

$$\lim_{x\to c;\ x\in X} (f(x))^n = L^n.$$

(B). [**5 Marks**] Let $g(x) \leq f(x) \leq h(x)$ for all $x \in X$. If $\lim_{x\to c;\ x\in X} g(x) = E = \lim_{x\to c;\ x\in X} h(x)$. Then prove that

$$\lim_{x\to c} f(x) = E.$$

Question 7 [10 Marks]. Consider the two functions $f : (0, \infty) \to \mathbb{R}$ and $g : [1, \infty) \to \mathbb{R}$ that are defined as $f(x) := \sqrt{x}$ and $g(x) := \sqrt{x}$ on their respective domains. Both functions have the same formula but different domains.

(A). [**6 Marks**] Prove that f is continuous at every $x_0 > 0$ and g is continuous at every $x_0 \geq 1$.

(B). [**2 Marks**] Fix $\varepsilon = 0.01$. Compute valid values of δ for $x_0 = 1$ and $x_0 = 4$, such that for all $x \in (0, \infty)$ and $|x - x_0| < \delta$, we have $|f(x) - f(x_0)| \leq \varepsilon$.

(C). [**2 Marks**] Fix $\varepsilon = 0.01$, can one find a single $\delta > 0$, independent of $x_0 \in [1, \infty)$, such that for all $x, x_0 \in [1, \infty)$ and $|x - x_0| < \delta$, we have $|g(x) - g(x_0)| \leq \varepsilon$.

Question 8 [10 Marks]. Answer the following questions.

(A). [**5 Marks**] Let $D \subset \mathbb{R}$ and $f : D \to \mathbb{R}$ is a function continuous on D. Let $c \in D$. If $f(c) \neq 0$, then there exists a $\delta > 0$ such that $\forall x \in D \cap (c - \delta, c + \delta)$, $f(x)$ keeps the same sign as $f(c)$. That is if $f(c) > 0$ (< 0), then there exists a $\delta > 0$ such that $\forall x \in D \cap (c - \delta, c + \delta)$, $f(x) > 0$ (< 0).

(B). [**5 Marks**] Let $f : [a, b] \to \mathbb{R}$ be a continuous function with $a < b$. If L is any real number satisfying

$$\min\{f(a), f(b)\} \leq L \leq \max\{f(a), f(b)\},$$

then there exists a point $c \in [a, b]$ such that $f(c) = L$.

Question 9 [10 Marks]. Let the real numbers a and b satisfy $a < b$. Let the function $f\colon [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Define

$$L(t) = f(a) + \frac{f(b) - f(a)}{b - a}(t - a), \qquad t \in [a, b],$$

and define $K(t) = f(t) - L(t),\ t \in [a, b]$.

(A). [**4 Marks**] Show that K satisfies the hypotheses of Rolle's theorem on $[a, b]$.

(B). [**6 Marks**] Define the function $\Phi(t) = K(t) - K(a + b - t),\ t \in [a, b]$ and show that there exists a point $c \in (a, b)$ for which

$$f'(c) + f'(a + b - c) = 2\frac{f(b) - f(a)}{b - a}.$$

Some hints.

These are suggestive and can be ignored. There are multiple ways to solve each question.
Q1. Introduce $d_n = x_{n+1} - x_n$. Use $x_{n+2} - x_n = d_{n+1} + d_n$ to relate limits and deduce $M = 2P$. Define $s_n = x_n + x_{n+1}$ and solve $2x_n = s_n - d_n$.
Q2A. Assume $a_n \to L$ and $a_n \to M$. The manipulate $|L - M|$.
Q2B. Rewrite

$$n\left(\frac{u_n}{u_{n+1}} - 1\right) \to L > 1.$$

Approximate $\frac{u_{n+1}}{u_n} \approx 1 - \frac{L}{n}$ and compare with a p-series.

Q3. Use the key identity: $3b_{2n} = a_{2n} + a_{2n+1} + a_{2n+2}$. Sum over $n = 1$ to N, rearrange sums into even/odd/shifted-even blocks, simplify, and let $N \to \infty$.
Q7. Use the identity

$$|\sqrt{x} - \sqrt{x_0}| = \frac{|x - x_0|}{\sqrt{x} + \sqrt{x_0}}$$

and appropriate lower bounds on the denominator depending on domain.
Q8. Sign preservation: choose $\varepsilon = \frac{|f(c)|}{2}$. Intermediate value theorem: let $E = \{x : f(x) < L\}$, set $c = \sup E$, and show $f(c) = L$.
Q9. Show $K(a) = K(b) = 0$, apply Rolle to K. Define $\Phi(t) = K(t) - K(a + b - t)$; use Rolle on Φ.

Cheat sheet [Informally correct statements]

Structure of Subsets of $\mathbb{R}$

Adherent points. x is adherent to X iff $\forall \varepsilon > 0$, $\exists y \in X$ with $|y - x| \leq \varepsilon$. Equivalently, every neighbourhood of x intersects X.
Limit points. x is a limit point of X iff $\forall \varepsilon > 0$, $\exists y \in X \setminus \{x\}$ with $|y - x| \leq \varepsilon$.
Closure. $\overline{X}$ = set of adherent points of X = X plus its limit points. X is closed iff $\overline{X} = X$.
Continuity
Definition f continuous at x_0 iff $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $|x - x_0| < \delta$ implies $|f(x) - f(x_0)| < \varepsilon$. Equivalent: use $\leq$ instead of $<$.
Sequential continuity. f continuous at x_0 iff $x_n \to x_0$ implies $f(x_n) \to f(x_0)$.
Differentiability
Definition f differentiable at x_0 if

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. Denote the limit $f'(x_0)$.
Chain Rule. If $f(x_0) = y_0$, f differentiable at x_0, g differentiable at y_0, then

$$(g \circ f)'(x_0) = g'(y_0)f'(x_0).$$

Derivative of x^α. For $f(x) = x^\alpha$, $x > 0$:

$$f'(x) = \alpha x^{\alpha-1}.$$

Rolle's theorem. Let $a < b$ be real numbers. Let $g : [a, b] \to \mathbb{R}$ be a continuous function on $[a, b]$ which is differentiable on (a, b) as well. Suppose also that $g(a) = g(b)$. Then there exists an $x \in (a, b)$ such that $g'(x) = 0$.
Extrema and Critical Points
Proposition (Local extrema are stationary). If f attains a local max or min at $x_0 \in (a, b)$ and is differentiable there, then $f'(x_0) = 0$.
Boundedness & Extreme Value
Lemma Continuous f on $[a, b]$ is bounded.
Proposition (Maximum principle). Continuous $f : [a, b] \to \mathbb{R}$ attains its max and min.
Sequences
Bolzano–Weierstrass theorem. Every bounded sequence has a convergent subsequence.
Convergence via even/odd subsequences. If $(a_{2n}) \to L$ and $(a_{2n+1}) \to L$, then $a_n \to L$. Reason: every n is even or odd, so eventually a_n lies in one of the two convergent subsequences.
limsup / liminf.

$$\limsup a_n = \inf_N \sup_{n \geq N} a_n, \qquad \liminf a_n = \sup_N \inf_{n \geq N} a_n.$$

Facts. (1). For $x > \limsup a_n$, eventually $a_n < x$. (2). For $y < \liminf a_n$, eventually $a_n > y$. (3). $\liminf a_n \leq \limsup a_n$. (4). Convergence $\iff$ $\liminf = \limsup = L$.
Series
If $\sum_{n=m}^{\infty} a_n = x$ and $\sum_{n=m}^{\infty} b_n = y$: (1). $\sum (a_n + b_n) = x + y$. (2). $\sum (ca_n) = cx$. (3). Shifts preserve convergence:

$$\sum_{n=m}^{\infty} a_n = \sum_{n=m}^{m+k-1} a_n + \sum_{n=m+k}^{\infty} a_n.$$

Useful Identities and Tricks
(1). $(1 + x)^r \geq 1 + r\,x$ for $r \geq 1$ and $x \geq -1$. (2). $2x_n = (x_n + x_{n+1}) - (x_{n+1} - x_n)$. (3). If $d_n = x_{n+1} - x_n$ and $d_n \to P$, then $d_n + d_{n+1} \to 2P$.